

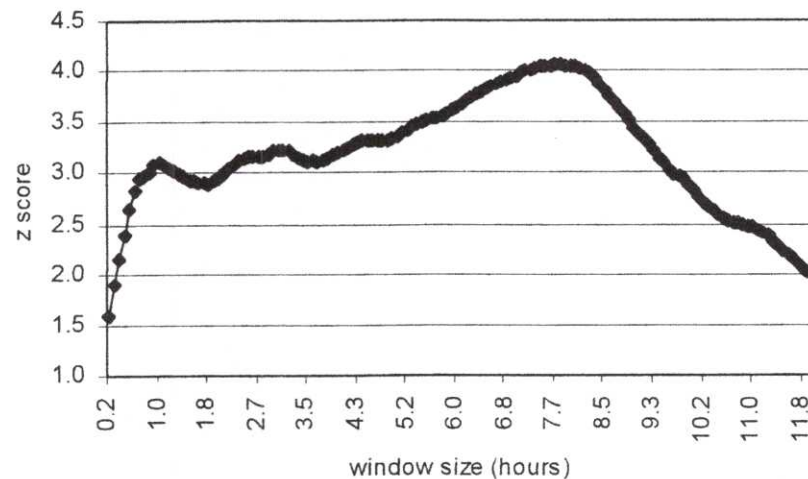


Fig. 8. Effect of window size length on differences between the distribution of all daily intercorrelation values vs. the distribution for September 11, in terms of a one-tailed z-score.

significantly positive: $r = 0.16$, $t(363 \text{ df}) = 3.08$, $p = 0.001$, one-tailed. If September 11 is removed from consideration: $r = 0.15$, $t(362 \text{ df}) = 2.88$, $p = 0.002$, one-tailed. And if all of the non-news days are removed (these are seen in Figure 9 as a column of points at 0 on the x-axis), the correlation remains significant: $r = 0.11$, $t(248 \text{ df}) = 1.76$, $p = 0.040$, one-tailed. A Kendall *tau* nonparametric correlation between the number of listed news events per day vs. the daily RNG intercorrelation value was also significant: $r = 0.062$, $n = 365$, $p = 0.037$, one-tailed.

Discussion

As mentioned above, one mundane explanation for the present results is that world events that captured mass human attention were associated with unusual surges of electrical power and use of telecommunications equipment, and this in turn might have created unusual environmental conditions that influenced the RNGs. While environmentally-induced artifacts are conceivable, there are four main arguments against this explanation: (1) in the case of September 11, the cross-RNG variance peaked over an hour before the terrorist events began to unfold, (2) the RNG intercorrelations reflected common changes among RNGs located around the world, (3) the RNGs are powered by voltage-regulated computer power supplies, and many PCs are further isolated from line power through surge suppressors and battery-powered, uninterruptible power supplies, and (4) the RNGs are designed to exclude first-order biases (i.e., drifts of the mean) through the use of XOR logic.

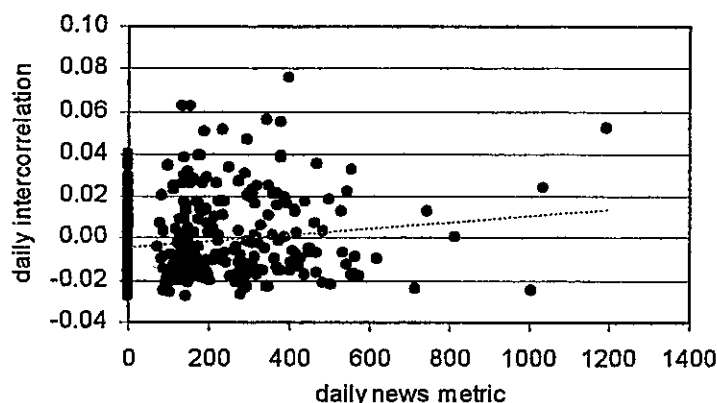


Fig. 9. Correlation between daily news metric and daily RNG intercorrelation values, $p = 0.001$, one-tailed. September 11, 2001, is associated with a news metric value of 398 in this graph.

These items argue against an artifactual explanation, but we can indirectly test the effects of electromagnetic interference on the RNGs by examining their outputs according to local clock time. That is, if the electromagnetic environment influenced the RNG circuits, then we would expect to see differences in RNG behavior between night and day. During the day, human use of electronic devices peaks, as does wide-spectrum electromagnetic noise, electric field strength, non-ionizing radiation, etc. During the night, all of these effects decline.

Figure 10 shows the z-score equivalent for variance across all RNGs, consolidated in 0.1-hour bins according to the local time of each RNG, over the entire month of September 2001. This graph summarizes 89.6 million 200-bit samples from all RNGs reporting in September 2001, for a total of 17.9 billion random bits. This provides enormous statistical power to identify diurnal influences, but no day-night differences or trends are observed: between 8:00 PM and 8:00 AM (night) and 8:00 AM and 8:00 PM (day), z (difference) = 0.53, $p = 0.30$, one-tailed. This provides no support for an electromagnetic artifact hypothesis.

Besides possible environmental artifacts and a global mind-matter interaction effect, what else might account for the observed results? One possibility is that these results might be due to chance. Another is that the results are due to a fortuitous choice of analysis methods. Follow-up tests with new data will help evaluate the viability of these possibilities.

Conclusion

Throughout history, philosophers have debated the perplexing, dualistic nature of subjective versus objective. In the 20th century, quantum theorists

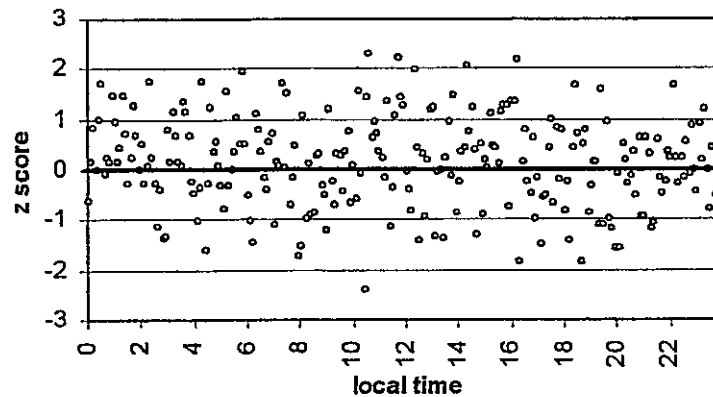


Fig. 10. RNG variance according to local time per RNG for the entire month of September 2001, in terms of z scores.

found themselves forced to seriously reconsider classical assumptions about observer vs. observed, and about mind vs. matter (Jahn, 1981; Jahn & Dunne, 1987; Stapp, 1999; Wilber, 1984). In the latter half of the 20th century, investigators developed increasingly rigorous methods for explicitly testing postulated mind-matter interactions (Radin & Nelson, 1989, in press). And as the 21st century begins, it appears that a cautious answer to the question used to taunt Bertrand Russell may be, "Yes, mind does matter." As for the observations discussed in this paper, whether they turn out to be a fluke due to the uncertainties of exploratory data analysis or something more interesting will be resolved by formalizing these analyses and testing them in future GCP data.

In sum, these analyses explored a new twist on the enduring riddle, "For whom does the bell toll?" The answer according to this analysis resonates with John Donne's words in the 16th century: "No man is an island. The bell tolls for thee."

Notes

- ¹ See the Web site <http://noosphere.princeton.edu> and Nelson (2001) for further details.
- ² More precisely, a normal distribution that approximates the underlying binomial distribution.
- ³ There is no easy answer for why the peak in this curve occurred before the terrorist attacks; the observable fact is that it did.

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